

# Self-replicating seedbox servers using programmable money

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01

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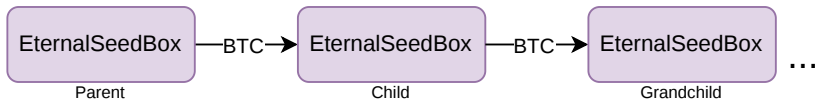
## Intro & Research Angle

**Content distribution without a human operator**

# Self-replicating infrastructure

Inspired by von Neumann's<sup>1</sup> theory of cellular automata, we set out to build the first self-replicating system that runs on **real** commercial hardware.

- **Self-replication:** the ability of an entity to autonomously reproduce.
- **In our case:** spawn identical clones that use the same software and run on bought hardware. Self-scaling network.



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<sup>1</sup>J. von Neumann (1966). *Theory of Self-Reproducing Automata*. Ed. by A. W. Burks. Urbana, IL: University of Illinois Press

## Use case

CDN's are operated by humans who serve the entity, rather than the content owners.

- Traditional CDN's concentrate operational control and content availability.
- We decentralize content, relinquish control, and replace humans with protocol.

To put it simply, we created an **alternative to YouTube's services**, by changing the delivery mechanism for Creative Commons content.

### From theory to real-world service

Self-replication is just the technical underpinning of the overarching goal: liberating mainstream content delivery networks.

## Just a P2P Network?

**Yes**, but it is complicated. We propose a network of **self-replicating seedbox servers**.

- Peer-to-peer networks distribute content across nodes, but the hardware is still managed by humans.
- We want to replace human oversight and control with **autonomous software agents**.

### The open research gap

Can the **infrastructure layer itself** be operator-independent, with no human controlling it after genesis?

# Self-replicating decentralized CDN infrastructure

We propose a network of **self-replicating seedbox servers** that:

- Run on commercial VPS instances.
- Rent the hardware themselves using Bitcoin.
- Seed Creative Commons YouTube content over BitTorrent.
- Can spawn children autonomously.
- Only human action is bootstrapping the genesis node.

**The research gap is filled – but can such a fleet actually *survive* and *grow* on its own?**

Democracy

SUPERORGANISM

ATTACK-RESILIENT SEEDBOX

Torrents

Fleet

Issues

My Issues

Voting History

AutoResearch

Settings

Create New Issue

AR

Alex Rivera

ID: 8821

Torrents

Total: 141 Healthy: 82 No Peers: 59 Exploding: 0

| Infohash        | Seeders | Leechers | Total Peer | Growth % | Shrink % | Exploding | Status  | Last Check          |
|-----------------|---------|----------|------------|----------|----------|-----------|---------|---------------------|
| 7d3398d3de9...  | 3       | 1        | 4          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-22 11:42:25 |
| b02e9158fcf8... | 3       | 1        | 4          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-22 11:38:40 |
| a4b2e628b8d...  | 3       | 2        | 5          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-15 12:37:35 |
| 991daf1e9bf8... | 2       | 1        | 3          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-22 11:59:33 |
| 5a4d845dbc6...  | 2       | 1        | 3          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-22 11:52:22 |
| ed56a0d49a0...  | 2       | 1        | 3          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-22 11:48:16 |
| 02ea9469271a... | 2       | 2        | 4          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-22 11:47:11 |
| 4fd7ffaaf47d... | 2       | 1        | 3          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-22 11:45:38 |
| 45d3b6e1e437... | 2       | 1        | 3          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-22 11:40:21 |
| 98475d456d87... | 2       | 2        | 4          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-15 13:32:53 |
| 5d52061f836b... | 2       | 6        | 8          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-15 12:57:38 |
| 63916d494a11... | 1       | 1        | 2          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-22 12:02:08 |
| bbabace72c59... | 1       | 3        | 4          | +0.0%    | -0.0%    | No        | Healthy | 2026-04-22 12:01:37 |

## Research question

Can a fleet of autonomous software agents, controlled only by inherited economic parameters, **survive** and **grow** without further human intervention?

**The main hypothesis:** network persistence and expansion according to specifications must be observed.

**Secondary hypothesis:** economically cautious nodes survive longer during financial downturn, and less-cautious nodes spawn more during financial excess.

# Research Novelty

We propose three contributions:

- a. **Engineering** — first self-replicating economic agent that buys its own infrastructure on an open market.
- b. **Testing Method** — unmodified production code run against a replica Bitcoin network and a mock VPS marketplace.
- c. **Empirical** — fleet dynamics and one heritable trait under selection.

**Research Novelty:** first self-replicating system on **real commercial infrastructure paid for with real money**.

02

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## EternalSeedBox

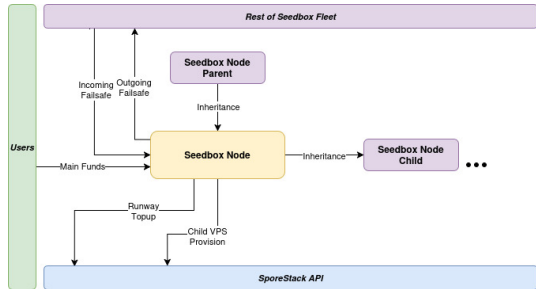
**How a node lives, reproduces, mutates, and dies**

# Single node overview

A node is a process running on a leased SporeStack VPS with:

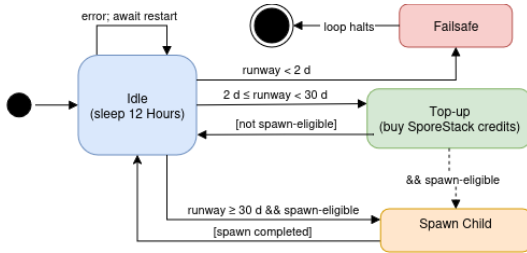
- **Bitcoin wallet**
- **Seedbox**
- **IPv8 overlay**
- **Decision loop**

**No central authority:** The fleet self-governs and no node has authority.



Arrows trace Bitcoin movements.

# The decision loop



Every 12 h, a node can:

- Failsafe** — runway < 2 days → sweep remaining funds to the richest peer.
- Top-up** — runway < 30 days → buy VPS credits to cover another 30 days
- Spawn** — if post-spawn runway enough → provision and fund a child.
- Idle** — wait for the next tick.

Identical code runs on every node.

# Reproduction mechanics

## Check post-spawn total runway

A node's **total runway** = VPS days + SporeStack credit + BTC. A node spawns only if it is left with  $\geq 60 \times (1 + c)$  days of runway after spawning, where  $c$  is the caution value.

- A child starts with a **30-day** pre-paid VPS and inherits **40%** of the parent's remaining Bitcoin.
- Actual Pipeline: mutate caution  $\rightarrow$  provision VPS  $\rightarrow$  fresh BTC wallet  $\rightarrow$  deploy code over SSH  $\rightarrow$  transfer inheritance.

Nodes must remain with significant runway **after** spawning a child.

## Caution trait

The single trait under selection.

Value  $c \in [0.35, 0.9]$ , with the genesis node starting at  $c = 0.5$ . Rest is mutated then inherited.

Caution scales the spawn threshold:

| caution $c$ | effective threshold |
|-------------|---------------------|
| 0.35        | 81 days             |
| 0.5         | 90 days             |
| 0.9         | 114 days            |

Low caution  $\rightarrow$  spawn sooner

High caution  $\rightarrow$  retain more

Mutation Formula:

$c_{parent} + \mathcal{N}(0, 0.05) + 0.2(0.5 - c_{parent}) \rightarrow$   
Gaussian drift with reversion to 0.5.

03

## Methodology

**Testing the node code offline and without cost**

# Faithful exercise of production code

## Unchanged node code

Core node code is executed **identically** to a real deployment. Only changes are time intervals (1 sim-day  $\approx$  22 s).

## Swapped the network layer only:

- Bitcoin  $\rightarrow$  private **regtest**, node reuses same paths.
- SporeStack API  $\rightarrow$  host-side **mock**, identical endpoints. Invoices paid in real regtest BTC.
- VPS  $\rightarrow$  local **Alpine LXC containers**. Parent deploys over the production SSH pipeline.
- Peer discovery via the same IPv8 tracker. A local IPv8 bootstrap server keeps the fleet offline.

**What runs locally is the code that would run on a live deployment.**

## Two experiments

### No-income run — closed system

- Genesis seeded with **only €10,000**.
- Every node spends only its **initial runway and inheritance**.

### Income run — open system

- Mock income pays the fleet
- Per-node income  $\approx$  invariant.
- Uneven daily shares.
- Income suspended at 60 nodes, resumed at 40  $\rightarrow$  alternate growth/drought.

### One caveat

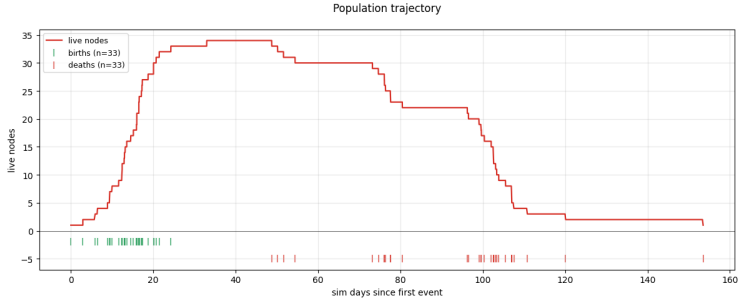
Under load, the regtest drops  $\sim$ half of all broadcasts, so income is set at  $\sim 10\times$  **per-node rent**. **The income run tests fleet mechanics under favorable conditions, not its economic viability.**

04

## Results

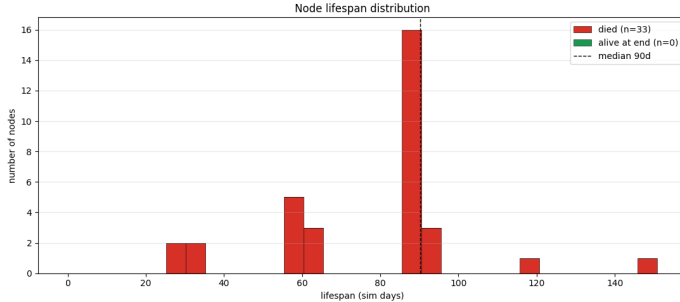
**Intended design confirmed, hypothesis tested**

# No-income run



- Every €3.27 buys one day, so the fleet's **total lifespan is 3061 node-days** (mean 93 days @ 33 nodes).
- One “old” survivor, as the failsafe concentrates residual funds into one node.

# No-income run

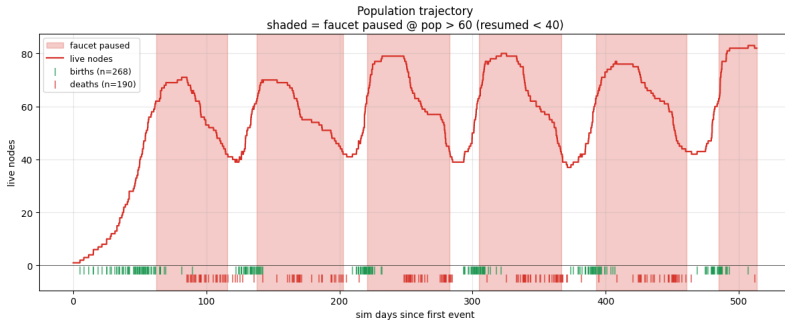


Each cluster is derived from a spawn constant: **90 d** is the post-spawn runway parents retain. **60 d** and **30 d** are the last two generations of children.

## Specification met

The cluster positions should be fixed beforehand, with the histogram confirming.

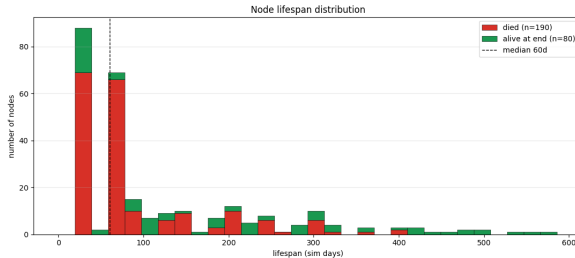
# Income run



- Income is cut at 60 nodes and restored at 40.
- The simulation ends when the host machine cannot hold the BTC chain in memory.

**Introducing income makes the fleet persist past 510 days.**

# Survival Distribution

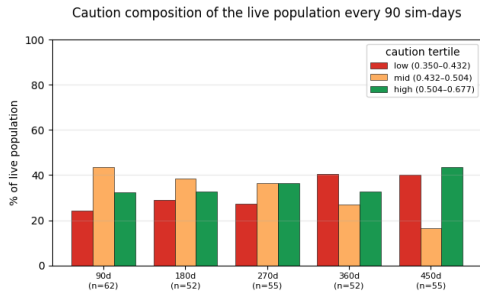


- Uneven early income splits the fleet, a node receiving more income early keeps large surplus, so **early wealth compounds** into a few long-lived nodes.

## High turnover

Survival only needs *some* node holding a surplus, so a well-funded minority eventually replaces the short-lived majority.

# Caution under selection: the hypothesis

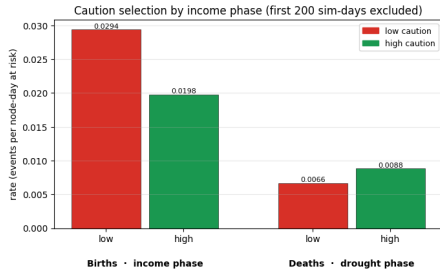


Caution should be **reverting to 0.5**. Observing the opposite means **selection pressure** is stronger than the artificial center-pull.

## Secondary Hypothesis

The income cycle rewards opposite strategies. Low caution exploits growth, high caution survives drought, while mid caution benefits from neither.

# Secondary hypothesis result



## Prediction 1 – confirmed

Low-caution reproduces ~48% more with income (0.0294 vs. 0.0198 births/day).

## Prediction 2 – disproved

High-caution has no survival advantage (0.0088 vs. 0.0066 deaths per day).

→ Income pauses were **too short** for extra caution to pay off. Testing on a more powerful host machine → future work.

05

## Discussion

**Fleet economics, and the limits of the evidence**

# What the runs establish

## Internal validity

- Production code works as specified.
- Replication verified under **both** the closed and income conditions.
- Fleet supports high node turnover.
- Genesis parameters are ultimately fully customizable.

## Limitations

- Forced node cap of 60, not economics.
- Regtest drops transactions, so node economy is inflated and uncertain.
- Single VPS provider.
- Revenue mechanism not yet designed.
- User content mechanism not designed.

**The mechanism is validated. Real-world viability remains to be.**

# Not cost-competitive

## Cost per TB-month

|                           |        |
|---------------------------|--------|
| EternalSeedBox            | €241   |
| Commercial object storage | €20–25 |
| Premium                   | ≈ 10×  |

Break-even **€3.27 / node / day** (€98 per 30 d, 400 GB served).

## Service becomes a deliberate niche

Each node pays for **retail VPS**, there is no economy of scale. The system buys availability that **survives takedown** and provides open-source accountability.

The market is **users whose content no one else will host**.

**Competitive only when content availability is the product.**

06

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## Conclusion

**Limitations, and what comes next**

# The answer

## Yes

The fleet survives and grows after genesis according to specification **when its income covers or exceeds the per-node rent.**

## Three contributions:

**Engineering** — first self-replicating economic agent that buys real infrastructure on an open market.

**Testing** — unmodified production code against Bitcoin and VPS-marketplace replicas.

**Empirical** — fleet dynamics and one heritable trait under selection.

# Thank you!

Questions?

## Future work

- **Payment incentive** — make consumers fund the nodes directly.
- **Decentralized moderation** — solve removal requests across a fleet with no central authority.
- **Longer drought, no node cap** — to test the unconfirmed half of the secondary hypothesis and to better observe large-scale fleet dynamics.

# Bibliography I



Neumann, J. von (1966). *Theory of Self-Reproducing Automata*. Ed. by A. W. Burks. Urbana, IL: University of Illinois Press.